



V93000 Specifications

EXA Scale Infrastructure

Version 7.10.3

Published on 2021-09-29

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Infrastructure specifications - V93000 EXA Scale systems

Scope of specifications

Advantest specifies and verifies the specifications at the DUT interface pogo level. Using an adequate DUT board, valid system calibration and a fixture delay measurement (TDR), these specifications are valid at the device under test as well. Specifications describe warranted product performance. Characteristics are included as typical values to provide additional useful information by describing typical non-warranted performance.

Test programming and debug

The SmarTest production test environment includes manufacturing test interfaces to operators, handling equipment and factory networks. A graphical testflow editor links device tests into a production-ready test program, where the tests are set up via fill-in-the-blank test functions. User-specific tests are programmed with test methods in Java. Links are available for design-to-test conversion. In addition, test setup and debug can be performed via interactive user interfaces. Result analysis tools are available for error locations, timing behavior, analog waveforms, shmoo plots, pin margin tests and memory bit map.

Supported channel cards

In addition, the following cards of Smart Scale systems are supported:

- AVI64
- DPS128
- DPS64
- PMUX
- PS9G
- PS1600
- PSSL
- UHC4T

These cards can be mixed within one card cage of the test head if the correct slot configuration is selected.

The following combinations in a single test program are not supported:

- PS1600 and PS5000
- DPS64 or DPS128 together with XPS

Compatibility

Software is compatible between all test heads.

The same DUT boards fit on all test heads. The docking system is identical for all test heads. The detailed docking pin positions might differ between different test heads. For details check the documentation on docking.

Operating system

Linux Red Hat Enterprise is used in all test head configurations.

Calibration and operation

Warm-up time	60 minutes
Basic maintenance period	6 months

Base calibration period (traceable calibration)	6 months
Calibration period (system auto adjustment)	3 months ^[1]
DC update period for enhanced accuracy of DC Scale AVI64	1 week

Further information

For more information about the Advantest V93000 SOC Series, please visit www.advantest.com.

Contact information

For more information about the Advantest V93000 SOC cards, please contact your local Advantest sales representative.

This information is subject to change without notice.

^[1] Valid at ambient temperature within ± 2.5 K of calibration temperature.

Test heads

Maximum channel count	CX test head	SX test head	LX test head
Number of card cages	1	2	3
Number of slots	9	18	27
PS1600	1152	2304	3456
DPS128	1152	2304	3456
XPS256	2304	4096 ^[2]	4096 ^[2]

The use of other digital channel cards, DC cards, analog instruments or RF cards may reduce the maximum channel count.

^[2] DUT interface limited (32 channel VHD pogo cable only, DUT interface has 128 pogo blocks available)

System slots

Supported slots

Maximum card count	CX test head	SX test head	LX test head
Number of card cages	1	2	3
Max. number of slots	9 ^[3]	18 ^[3]	27 ^[3]

Note

Mixing of different card types is supported within one test head. DC/DC power for V93000 EXA Scale systems is different than for Smart Scale and Pin Scale systems. This means you need to configure the system with the correct power to operate correctly.

Refer to the corresponding card specification for details.

^[3] No Utility Card installed, using Internal Utility Line option.

Smart Scale device power supplies

Supported channel slot device power supplies

Maximum channel count	CX test head	SX test head	LX test head
Number of slots	9	18	27
DC Scale DPS128-HC (E8023CS)	1152	2048 ^[4]	2048 ^[4]
DC Scale DPS64-HC (E8023CSH)	576	1152	1728
DC Scale UHC4T (E8024UH)	36	72	108
DC Scale XPS256	2304	4096 ^[5]	4096 ^[5]

Note

Mixing different device power supply types within one Test Head is supported.
Refer to DC Scale device power supply specification sheet for more information.

^[4] DUT interface limited (16 channel HD pogo blocks, DUT interface has 128 pogo blocks available)

^[5] DUT interface limited (32 channel VHD pogo cable assembly only, DUT interface has 128 pogo blocks available)

Parametric measurements

Direct DVM Access (at utility pogo block)

A DVM can be routed through the DC rail onto each tester channel.

Supported Digital Multimeter (DMM)

Keysight 3458A with option 002

Absolute voltage measurement accuracy

$\pm 40\mu\text{V}$ + DMM specification

Input voltage range

-5 V to 7.5 V

Utility resources

Utility resources are the utility power supplies and the utility lines that you can use to supply and control additional hardware on the DUT board.

Utility lines of V93000 EXA Scale systems allow you to control external devices such as relays, solenoids or indicators, depending on the application. The lines are implemented as FET switch outputs.

The utility power supplies of V93000 EXA Scale systems support DUT boards with additional active components. Typical examples for these active components are relays, switches, operational amplifiers, and buffers.

Internal utility resources

Internal utility resources of a V93000 EXA Scale system in the basic configuration consist of 32 statically controlled utility lines, a +5 V utility power supply and the Internal Utility Power Option. For the SX and LX test heads, the internal utility resources can optionally be extended by 32 more utility lines and one +5 V supply. Internal utility resources can be replaced and extended by an optional Utility Card.

Internal utility lines

In the following table, only the "Basic configuration" columns with "Optional Utility Card (EK051A)" set to "none" apply to the internal utility lines. The other columns require the installation of minimum one optional *Utility Card (EK051 and EK051A)* on page 13 that replaces the internal utility lines.

Configuration: Basic configuration		Utility Card configuration		
TH: CX / SX / LX		CX / SX / LX		
Slots: 9 / 18 / 27		9 / 18 / 27		
CX test head supports only one internal utility module.				
Internal utility modules: 1	2 ^[6]	none	none	
Optional Utility Cards (EK051 and EK051A): none	none	1	2	
Configuration				
Number of utility lines	32	64	256	512

^[6] Option of the SX and LX test heads.

Configuration: Basic configuration			Utility Card configuration	
TH: CX / SX / LX			CX / SX / LX	
Slots: 9 / 18 / 27			9 / 18 / 27	
CX test head supports only one internal utility module.				
Internal utility modules: 1		2 ^[6]	none	none
Optional Utility Cards (EK051 and EK051A): none		none	1	2
Pogo block	UTIL 1 (165)	UTIL 1 (165)	UTIL 1 ... 8 (165 ... 172)	UTIL 1 ... 8 (165 ... 172)
		UTIL 3 (167)		DPS 1 ... 8 (265 ... 272)
			Note: The 4-row utility pogo blocks of the second Utility Card replace all existing 3-row DPS-slot pogo blocks. The DPS block sequence of the second Utility Card differs from the UTIL block sequence of the first card. For the correct pogo block assignment of the second card, refer to "Utility pogo block mapping" in TDC topic "Utility pogo block - Utility Card (V93000 EXA Scale systems)" (347642).	
Sequencer control	no	no	yes	yes
Output				
Type	Open drain			
Maximum current	100 mA per line			
Cut-off line current	130 ±15 mA after 250 µs			
Voltage				
Output switch open, no load applied		Each utility line has an internal pull-up resistor of 10 kΩ to the internal supply voltage of 3.3 V. We recommend using external pull up resistors for applications with different requirements.		
Output switch closed				
V _{ds}		0.5 V maximum		
R _{on}		< 5 Ω		
Input				
Type	TTL			
Voltage range	0 V ... 5.0 V			
Maximum voltage rating	5.5 V, internal clamp diode to 5 V			
Maximum clamp current				
per channel	200 mA	200 mA	200 mA	200 mA

^[6] Option of the SX and LX test heads.

Configuration: Basic configuration			Utility Card configuration	
TH: CX / SX / LX			CX / SX / LX	
Slots: 9 / 18 / 27			9 / 18 / 27	
CX test head supports only one internal utility module.				
Internal utility modules: 1		2 ^[6]	none	none
Optional Utility Cards (EK051 and EK051A): none		none	1	2
total	800 mA	800 mA per utility module	800 mA	800 mA per card
Input impedance		10 kΩ typ.		
Input thresholds				
Maximum V_{IL}		1.3 V typ.		
Minimum V_{IH}		2.1 V typ.		

Internal utility power supplies

Utility power supply - UP5V pins

In the following table, only the "Basic configuration" columns with "Optional Utility Cards (EK051A)" set to "none" apply to the internal UP5V utility power supply. The other columns require the installation of minimum one optional *Utility Card (EK051 and EK051A)* on page 13 that replaces the internal UP5V utility power supply.

Configuration: Basic configuration			Utility Card configuration	
TH: CX / SX / LX			CX / SX / LX	
Slots: 9 / 18 / 27			9 / 18 / 27	
CX test head supports only one internal utility module.				
Internal utility modules: 1		2 ^[7]	none	none
Optional Utility Cards (EK051 and EK051A): none		none	1	2
Configuration				
Total number of supplies	1	2	1	2

^[6] Option of the SX and LX test heads.

^[7] Option of the SX and LX test heads.

Configuration: Basic configuration			Utility Card configuration	
TH: CX / SX / LX			CX / SX / LX	
Slots: 9 / 18 / 27			9 / 18 / 27	
CX test head supports only one internal utility module.				
Internal utility modules: 1		2 ^[7]	none	none
Optional Utility Cards (EK051 and EK051A): none		none	1	2
Supply routed to pogo block	UTIL 1 (165)	UTIL 1 (165)	UTIL 1 ... 8 (165 ... 172)	UTIL 1 ... 8 (165 ... 172)
		UTIL 3 (167)		DPS 1 ... 8 (265 ... 272)
			Note: The 4-row utility pogo blocks of the second Utility Card replace all existing 3-row DPS-slot pogo blocks. The DPS block sequence of the second Utility Card differs from the UTIL block sequence of the first card. For the correct pogo block assignment of the second card, refer to "Utility pogo block mapping" in TDC topic "Utility pogo block - Utility Card (V93000 EXA Scale systems)" (347642).	

Characteristics per supply

Output voltage		+5.0 V $\pm$ 2 % sensed to the pogo pin		
Total output current		3.5 A	7.0 A	8.0 A
Maximum load capacitance		10000 μ F		
Pogo pins	+5.0 V:	UP5V		
	GND:	UGND		
		Note: UP5V pins of different card cages or Utility Cards must not be connected together on the DUT board.		

Internal Utility Power Option

The Internal Utility Power Option can be routed to a 3-row DPS-slot pogo block if maximum one optional Utility card (EK051 or EK051A) is installed. The supply provides compatibility with the legacy SE-UDPS and requires the EK051UP pogo cable assembly.

^[7] Option of the SX and LX test heads.

Configuration

TH: CX / SX / LX	
Slots: 9 / 18 / 27	
Optional Utility Cards (EK051 and EK051A): = 1^[8]	
Total number of supplies	1
Supply routed to pogo block	DPS 8 (268) ^[9]

Characteristics

Voltage	Accuracy	Maximum current	Pogo pins
			The "Internal Utility Power Option" is routed to a 3-row DPS pogo block.
+5.0 V, sensed	±2 %	3 A	P1, PS1, G1 (two pins)
+12.0 V, sensed	±2 %	1 A	P2, PS2, G2 (two pins)
+15.0 V, sensed	±2 %	1 A	P3, PS3, G3 (two pins)
-15.0 V, sensed	±2 %	1 A	P4, PS4, G4 (two pins)

Utility Card (EK051 and EK051A)

The Utility Card provides 256 utility lines and several utility power supplies on eight 4-row pogo blocks. The cards can reside in any slot of a V93000 EXA Scale system. There are two variants available, EK051 and EK051A. The EK051 Utility card is used in SmarTest 8 only. The EK051A Utility card is used in both SmarTest 7 (SmarTest version 7.10 and higher) and SmarTest 8.

Utility lines

In the following table, only the "Utility Card configuration" columns with "Internal utility module" set to "none" apply to the Utility Card. The other columns show the capabilities of the *Internal utility resources* on page 9 with no Utility Card installed.

Configuration: Basic configuration		Utility Card configuration	
TH: CX / SX / LX		CX / SX / LX	
Slots: 9 / 18 / 27		9 / 18 / 27	
CX test head supports only one internal utility module.			
Internal utility modules: 1	2 ^[10]	none	none
Optional Utility Cards (EK051 and EK051A): none	none	1	2

Configuration

Number of utility lines	32	64	256	512
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^[8] Maximum one Utility card (EK051 or EK051A) can be installed for accessing the Internal Utility Power Option. The second Utility card occupies the pogo block position that is required for the Internal Utility Power Option.

^[9] Pogo block DPS 8 (268) is the recommended position as defined in the V93000 Standard Configuration Guidelines.

^[10] Option of the SX and LX test heads.

Configuration: Basic configuration			Utility Card configuration	
TH: CX / SX / LX			CX / SX / LX	
Slots: 9 / 18 / 27			9 / 18 / 27	
CX test head supports only one internal utility module.				
Internal utility modules: 1		2 ^[10]	none	none
Optional Utility Cards (EK051 and EK051A): none		none	1	2
Pogo block	UTIL 1 (165)	UTIL 1 (165) UTIL 3 (167)	UTIL 1 ... 8 (165 ... 172)	UTIL 1 ... 8 (165 ... 172) DPS 1 ... 8 (265 ... 272)
			Note: The 4-row utility pogo blocks of the second Utility Card replace all existing 3-row DPS-slot pogo blocks. The DPS block sequence of the second Utility Card differs from the UTIL block sequence of the first card. For the correct pogo block assignment of the second card, refer to "Utility pogo block mapping" in TDC topic "Utility pogo block - Utility Card (V93000 EXA Scale systems)" (347642).	
Sequencer control	no	no	yes	yes
Output				
Type	Open drain			
Maximum current	100 mA per line			
Cut-off line current	130 ±15 mA after 250 µs			
Voltage				
Output switch open, no load applied		Each utility line has an internal pull-up resistor of 10 kΩ to the internal supply voltage of 3.3 V. We recommend using external pull up resistors for applications with different requirements.		
Output switch closed				
V _{ds}		0.5 V maximum		
R _{on}		< 5 Ω		
Input				
Type	TTL			
Voltage range	0 V ... 5.0 V			
Maximum voltage rating	5.5 V, internal clamp diode to 5 V			
Maximum clamp current				
per channel	200 mA	200 mA	200 mA	200 mA

^[10] Option of the SX and LX test heads.

Configuration: Basic configuration			Utility Card configuration	
TH: CX / SX / LX			CX / SX / LX	
Slots: 9 / 18 / 27			9 / 18 / 27	
CX test head supports only one internal utility module.				
Internal utility modules: 1		2 ^[10]	none	none
Optional Utility Cards (EK051 and EK051A): none		none	1	2
total	800 mA	800 mA per utility module	800 mA	800 mA per card
Input impedance		10 kΩ typ.		
Input thresholds				
Maximum V_{IL}		1.3 V typ.		
Minimum V_{IH}		2.1 V typ.		

Utility power supplies

Utility power supply - UP5V pins

In the following table, only the "Utility Card configuration" columns with "Internal utility module" set to "none" apply to the Utility Card. The other columns show the capabilities of the *Internal utility resources* on page 9 with no Utility Card installed.

Configuration: Basic configuration			Utility Card configuration	
TH: CX / SX / LX			CX / SX / LX	
Slots: 9 / 18 / 27			9 / 18 / 27	
CX test head supports only one internal utility module.				
Internal utility modules: 1		2 ^[11]	none	none
Optional Utility Cards (EK051 and EK051A): none		none	1	2
Configuration				
Total number of supplies	1	2	1	2

^[10] Option of the SX and LX test heads.

^[11] Option of the SX and LX test heads.

Configuration: Basic configuration			Utility Card configuration	
TH: CX / SX / LX			CX / SX / LX	
Slots: 9 / 18 / 27			9 / 18 / 27	
CX test head supports only one internal utility module.				
Internal utility modules: 1		2 ^[11]	none	none
Optional Utility Cards (EK051 and EK051A): none		none	1	2
Supply routed to pogo block	UTIL 1 (165)	UTIL 1 (165)	UTIL 1 ... 8 (165 ... 172)	UTIL 1 ... 8 (165 ... 172)
		UTIL 3 (167)		DPS 1 ... 8 (265 ... 272)
			Note: The 4-row utility pogo blocks of the second Utility Card replace all existing 3-row DPS-slot pogo blocks. The DPS block sequence of the second Utility Card differs from the UTIL block sequence of the first card. For the correct pogo block assignment of the second card, refer to "Utility pogo block mapping" in TDC topic "Utility pogo block - Utility Card (V93000 EXA Scale systems)" (347642).	

Characteristics per supply

Output voltage		+5.0 V ±2 % sensed to the pogo pin			
Total output current		3.5 A	7.0 A	8.0 A	16.0 A
Maximum load capacitance		10000 μF			
Pogo pins	+5.0 V:	UP5V			
	GND:	UGND			
		Note: UP5V pins of different card cages or Utility Cards must not be connected together on the DUT board.			

Utility power supply - XP pins

^[11] Option of the SX and LX test heads.

Configuration

TH: CX / SX / LX		
Slots: 9 / 18 / 27		
Utility Cards (EK051 and EK051A):	1 ^[12]	2
Total number of supplies	1	2
Supply routed to pogo block	UTIL 1 ... 8 (165 ... 172)	UTIL 1 ... 8 (165 ... 172) DPS 1 ... 8 (265 ... 272)
Note: The 4-row utility pogo blocks of the second Utility Card replace all existing 3-row DPS-slot pogo blocks. The DPS block sequence of the second Utility Card differs from the UTIL block sequence of the first card. For the correct pogo block assignment of the second card, refer to "Utility pogo block mapping" in TDC topic "Utility pogo block - Utility Card (V93000 EXA Scale systems)" (347642).		

Characteristics per power supply

Voltage	Accuracy	Maximum current	Max. load capacitance	Label
+5.0 V, sensed	±2 %	12.8 A	10000 µF	XPP5V
-5.0 V, sensed	±2 %	6 A	7000 µF	XPM5V
+3.3 V, sensed	±2 %	6 A	10000 µF	XPP3V3

Utility power supply - UPP15V / UPM15V / F24V pins

Configuration

TH: CX / SX / LX			
Slots: 9 / 18 / 27			
Utility Cards (EK051 and EK051A): 1 ^[13]			2
Total number of supplies		1	2
Supply routed to pogo block			
Voltage	Label		
+15 V	UPP15V	UTIL 2 + 8 (166 + 172)	UTIL 2 + 8 (166 + 172) DPS 1 + 7 (266 + 272)
-15 V	UPM15V	UTIL 3 + 5 (167 + 169)	UTIL 3 + 5 (167 + 169) DPS 2 + 4 (267 + 269)

^[12] At least one Utility card (EK051 or EK051A) must be installed if the utility power supply on XP pins is to be used.

^[13] At least one Utility card (EK051 or EK051A) must be installed if the utility power supplies on UPP15V / UPM15V / F24V pins are to be used.

TH: CX / SX / LX			
Slots: 9 / 18 / 27			
Utility Cards (EK051 and EK051A):		1 ^[13]	2
24 V floating ^[14]	F24VA	UTIL 4 (168)	UTIL 4 (168) DPS 8 (268)
	F24VB	UTIL 6 (170)	UTIL 6 (170) DPS 5 (270)
Note: The 4-row utility pogo blocks of the second Utility Card replace all existing 3-row DPS-slot pogo blocks. The DPS block sequence of the second Utility Card differs from the UTIL block sequence of the first card. For the correct pogo block assignment of the second card, refer to "Utility pogo block mapping" in TDC topic "Utility pogo block - Utility Card (V93000 EXA Scale systems)" (347642).			

Characteristics per power supply

Voltage	Accuracy	Maximum current	Max. load capacitance	Label
+15.0 V, sensed	±2 %	3 A	1000 µF	UPP15V
-15.0 V, sensed	±2 %	3 A	1000 µF	UPM15V
+24.0 V, sensed, floating ^[14]	±4 %	2 * 1 A	200 µF	F24VA and F24VB

Note

Connect all pogo pins (of all blocks) of a supply domain to a separate power plane on the DUT board. Connect all sense pins (of all blocks) of a supply domain to the power plane.

Maximum offset of utility ground to system ground: ±0.5 V

▲ CAUTION

Utility power supply pogo pins of utility blocks that are connected to different internal utility modules or Utility Cards must not be connected together.

^[13] At least one Utility card (EK051 or EK051A) must be installed if the utility power supplies on UPP15V / UPM15V / F24V pins are to be used.

^[14] Floating range: ±24 V max.

Miscellaneous characteristics

Side panel port

Characteristics	CX test head	SX test head	LX test head
Number of card cages	1	2	3
DC rails	1	1	1

Reference 10MHz

Frequency	10 MHz
Impedance	50 Ω
Level	6 dBm (1.26Vpp) ^[15]
	1.26 Vpp
	AC-coupled
Accuracy	1 ppm ^[16]

DUT interface types

All test heads support the enhanced DUT Scale interface and feature the "infinite plane" concept, which means any DUT board size fits on any DUT interface.

^[15] Into 50 Ω

^[16] Without external adjustment

Environmental

Operating	15 °C to 30 °C (59 °F to 86 °F)
Maximum rate of change	± 2 #/h (3.6 #/h)
Specification warranted temperature	20 °C to 30 °C (68 °F to 86 °F)
Maximum humidity at 30°C	< 70 % R.H. ^[17] , non-condensing
Storage (without water)	-40 °C to +70 °C

^[17] Relative humidity

Power requirements

Maximum power consumption

Test head type	CX test head	SX test head	LX test head
Number of card cages	1	2	3
Number of slots	9	18	27
Maximum power	20 kW	40 kW	56 kW

Cooling requirements

For use with one CX (9 slot) test head

Cooling unit E2760FAE Small liquid-liquid cooling unit

Cooling capability	18 kW @ 30 °C regulated secondary water temperature
Cooling medium	De-ionized water
Primary chilled water temperature range	5 °C to 20 °C (41 °F to 68 °F)

For use with one SX (18 slot) test head

Cooling unit E2760FAT Liquid-liquid cooling unit

Cooling capability	36 kW @ 30°C regulated secondary water temperature
Cooling medium	De-ionized water
Primary chilled water temperature range	5 °C to 20 °C (41 °F to 68 °F)

For use with one LX (27 slot) test head

Cooling unit E2760FAX and E2760FAXV Liquid-liquid cooling unit

Cooling capability	54 kW @ 30°C regulated secondary water temperature
Cooling medium	De-ionized water
Primary chilled water temperature range	5 °C to 20 °C (41 °F to 68 °F)

Weights and dimensions

Test system weight with manipulator and support rack (fully loaded)

CX test head - 1 CC	< 1300 kg
SX test head - 2 CC	< 1500 kg
LX test head - 3 CC	< 2300 kg

Test head dimensions (height x width x depth, incl. docking)

CX test head - 1 CC	0.65 m x 0.73 m x 0.78 m
SX test head - 2 CC	0.65 m x 0.73 m x 0.92 m
LX test head - 3 CC	0.67 m x 0.97 m x 1.02 m

Dimensions of the test cell (TH installed to the manipulator) (height x width x depth, incl. docking)

CX test head - 1 CC	1.92 m x 1.45 m x 2.6 m
SX test head - 2 CC	2.23 m x 1.63 m x 2.67 m
LX test head - 3 CC	2 m x 1.36 m x 2.81 m

Cooling unit E2760FAE

Dimensions (height x width x depth)	0.832 m (h) x 0.363 m (w) x 0.815 m (d)
Net weight	81 kg

Cooling unit E2760FAT

Dimensions (height x width x depth)	0.852 m (h) x 0.443 m (w) x 0.815 m (d)
Net weight	92 kg

Cooling unit E2760FAX

Dimensions (height x width x depth)	1.095 m (h) x 0.504 m (w) x 1.078 m (d)
Net weight	173 kg

Cooling unit E2760FAXV

Dimensions (height x width x depth)	1.095 m (h) x 0.504 m (w) x 1.078 m (d)
Net weight	180 kg